Ordered fields, supremum/infimum and the completeness axiom

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Properties R_1 and R_2

Introduction: Properties R_1 and R_2

- We now begin to study more closely the **structure of the number sets** that we introduced earlier, and in particular: the set $\mathbb{Q}$ of **rational numbers** and the set $\mathbb{R}$ of **real numbers**.
- In this section we see the properties of the operations of **sum** (R_1) and **product** (R_2)
- From them follows the possibility of performing the four fundamental operations

Property R_1: the sum

We denote by a, b, c three generic **rational** or **real** numbers.

$R_1 \rightarrow$ In $\mathbb{Q}$ and $\mathbb{R}$ an operation is defined (called **addition** or **sum**) which has the following four properties:

1. $\forall a, b, \quad a + b = b + a \qquad$ (**commutative property**)
2. $\forall a, b, c, \quad (a + b) + c = a + (b + c) \qquad$ (**associative property**)
3. there exists a **neutral element for the sum**, denoted by 0, such that:
$$\forall a, \quad a + 0 = a$$
4. for every a there exists an **inverse** element of a **with respect to the sum**, called the **opposite** of a and denoted by $-a$, such that:
$$a + (-a) = 0$$

Property R_2: the product

$R_2 \rightarrow$ In $\mathbb{Q}$ and $\mathbb{R}$ an operation is defined (called **multiplication** or **product**) which has the following four properties:

1. $\forall a, b, \quad a \cdot b = b \cdot a$ (**commutative property**)
2. $\forall a, b, c, \quad (a \cdot b) \cdot c = a \cdot (b \cdot c)$ (**associative property**)
3. there exists a **neutral element for the product**, denoted by 1, such that:
$$\forall a, \quad a \cdot 1 = a$$
4. for every $a \neq 0$ there exists an **inverse** element of a **with respect to the product**, called the **reciprocal** of a and denoted by a^{-1}, such that:
$$a \cdot a^{-1} = 1$$

Distributive property and the four operations

- The operations of sum and product are linked by the following property:

$$\forall a, b, c, \qquad (a+b)\cdot c = a\cdot c + b\cdot c \qquad (\textbf{distributive property})$$

- From properties R_1 and R_2 follows the **possibility of performing the four fundamental operations without restrictions**:
  1. **addition** and **multiplication** (defined above)
  2. **subtraction**, by setting:

$$a - b = a + (-b)$$

  3. **division**, by setting:

$$a/b = a \cdot b^{-1} \quad \text{provided that } b \neq 0$$

Summary

- R_1: the sum is commutative, associative, has neutral element 0 and every a has the opposite $-a$
- R_2: the product is commutative, associative, has neutral element 1 and every $a \neq 0$ has the reciprocal a^{-1}
- Distributive property: $(a + b) \cdot c = a \cdot c + b \cdot c$
- In $\mathbb{Q}$ and $\mathbb{R}$ sum, product, subtraction and division (for $b \neq 0$) can be performed without restrictions

Commensurable magnitudes

Introduction: Commensurable magnitudes

- In this section we introduce geometric **magnitudes** and how to compare them
- We define **homogeneous**, **commensurable** and **incommensurable** magnitudes
- Incommensurable magnitudes will show that rational numbers are not enough to measure everything

Homogeneous magnitudes

In mathematics, the word **magnitude** denotes a **property** of a **geometric figure** that can be **measured**. Examples of magnitudes are: the length of a segment, the measure of an angle, the area of a plane figure or the volume of a solid.

Definition 1: of homogeneous magnitudes

*Two magnitudes are **homogeneous** if they have the same dimension, i.e., if they can be expressed using the same unit of measurement.*

- The length of a segment and the height of a solid are two homogeneous magnitudes; indeed, both can be expressed with the same unit of length.
- The area of a square and the volume of a cone are not homogeneous magnitudes; indeed, area is expressed in square meters or in other units of surface, which however cannot be used to express volume.

Commensurable and incommensurable magnitudes

Definition 2: of commensurable and incommensurable magnitudes

Two homogeneous magnitudes a and b are called ***commensurable*** *if their ratio is a rational number, i.e., if $a/b \in \mathbb{Q}$. They are instead called* ***incommensurable*** *if their ratio is not a rational number, i.e., if $a/b \notin \mathbb{Q}$.*

Example

Commensurable magnitudes. The volume of a sphere and the volume of the cylinder circumscribed about the sphere are commensurable. In formulas, the volume V_s of the sphere as a function of its radius r is given by the formula:

$$V_s = \frac{4}{3}\,\pi\,r^3$$

The cylinder circumscribed about it has base radius equal to the radius r of the sphere and height h equal to twice the radius. The volume V_c of this cylinder is therefore given by the formula:

$$V_c = \pi\,r^2\,h = \pi\,r^2\,2\,r = 2\,\pi\,r^3$$

Taking the ratio we obtain:

$$\frac{V_s}{V_c} = \frac{\frac{4}{3}\,\pi\,r^3}{2\,\pi\,r^3} = \frac{2}{3} \in \mathbb{Q}$$

Example

Incommensurable magnitudes. The length of a circle and the length of its diameter are incommensurable. In formulas, denoting by d the diameter of the circle and by c the length of the circle, we have:

$$c = \pi\, d$$

Now, taking the ratio between the length of the circle and its diameter, we obtain:

$$\frac{c}{d} = \frac{\pi\, d}{d} = \pi \notin \mathbb{Q}$$

A proof that $\pi \notin \mathbb{Q}$ will be given later.

Summary

- A **magnitude** is a measurable property of a geometric figure
- **Homogeneous** magnitudes: expressed with the same unit of measurement
- **Commensurable**: rational ratio (sphere and circumscribed cylinder: $\frac{2}{3}$)
- **Incommensurable**: non-rational ratio (circle and diameter: π)

Geometric representation of $\mathbb{Q}$

Introduction: Geometric representation of $\mathbb{Q}$

- In this section we represent rational numbers as points of a line
- Once a unit of measurement is fixed, rational numbers correspond to the points identifying segments commensurable with it

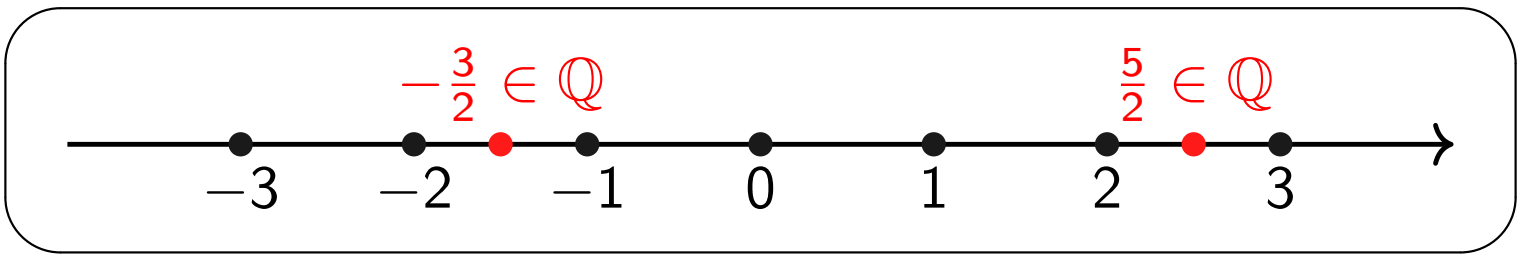
The Euclidean line

A geometric representation of $\mathbb{Q}$ can be obtained by associating with each rational number a point of the **Euclidean line** (which can be imagined as a line in the plane of infinite length)

- With an arbitrarily chosen point of the Euclidean line we associate 0, and with another point, distinct from the first, we associate 1, thus identifying the oriented segment 01, which constitutes the **unit of measurement**
- At this point we have a **one-to-one correspondence** between the rational numbers and those points P of the line that are endpoints of oriented segments $0P$ **commensurable** with 01. Indeed, if the rational number p (its **abscissa**) is associated with the point P, the segment $0P$ measures p times the unit segment 01, and therefore the ratio of the two lengths is:

$$\frac{0P}{01} = \frac{p}{1} = p \in \mathbb{Q}$$

The Euclidean line



Total order relations

Introduction: Total order relations

- In this section we recall that "$\leq$" is a total order relation on $\mathbb{Q}$ and on $\mathbb{R}$
- This allows us to compare any two numbers, the basis for defining maxima, minima and suprema

$\mathbb{Q}$ and $\mathbb{R}$ are totally ordered

Observation 1

The sets of rational numbers $\mathbb{Q}$ and of real numbers $\mathbb{R}$ with the relation "less than or equal to" ("$\leq$") are totally ordered sets.

- The relations "less than or equal to" ("$\leq$") for the set of rational or real numbers:

$$R_{\leq} = \left\{ (a, b) : a, b \in \mathbb{Q} \quad \text{and} \quad a \leq b \right\}$$

$$R_{\leq} = \left\{ (a, b) : a, b \in \mathbb{R} \quad \text{and} \quad a \leq b \right\}$$

are **partial order relations**.

Partial and total order

That is, they satisfy the following properties:

$$\begin{aligned}
&\forall a, && a \leq a && && && \text{(\textbf{reflexive} property)}\\
&\forall a, b, && a \leq b && \text{and} && b \leq a \;\Rightarrow\; a = b && \text{(\textbf{antisymmetric} property)}\\
&\forall a, b, c, && a \leq b && \text{and} && b \leq c \;\Rightarrow\; a \leq c && \text{(\textbf{transitive} property)}
\end{aligned}$$

Moreover, the relations "less than or equal to" ("$\leq$") for $\mathbb{Q}$ and $\mathbb{R}$ are **total relations** since:

$$\forall a, b, \qquad a \leq b \quad \text{or} \quad b \leq a$$

- Consequently, the pair consisting of the set $\mathbb{Q}$ or $\mathbb{R}$ and the corresponding relation $R_{\leq}$ is a totally ordered set.

The trichotomy property

Observation 2: trichotomy property

For every pair of numbers a, b (rational or real) ***exactly one*** *of the three relations holds:*

$$a < b, \qquad a = b, \qquad a > b$$

- This is not a new property: it is a consequence of the fact that "$\leq$" is a **total** order relation, and for this reason we do not count it among the axioms R_1, R_2, R_3 and R_4

Proof

Recall that $a < b$ means $a \leq b$ and $a \neq b$.

At least one of the three relations holds. Since the relation is total, we have $a \leq b$ or $b \leq a$:

- if $a \leq b$ and $a \neq b$, then $a < b$;
- if $b \leq a$ and $a \neq b$, then $a > b$;
- if $a = b$ the second relation holds.

At most one of the three relations holds. The relations $a < b$ and $a = b$ cannot hold together, because the first one requires $a \neq b$; the same for $a > b$ and $a = b$. Finally, if $a < b$ and $a > b$ held together, we would have $a \leq b$ and $b \leq a$, hence $a = b$ by the antisymmetric property, contradicting $a \neq b$. □

Property R_3

Introduction: Property R_3

- In this section we highlight the properties of the ordering of rational or real numbers
- Property R_3 links the order to the operations of sum and product
- All the rules of algebraic computation follow from R_1, R_2, R_3

Property R_3

$R_3 \rightarrow$ In $\mathbb{Q}$ and $\mathbb{R}$ the total order relation "less than or equal to" ("$\leq$") is defined, compatible with the algebraic structure[1], that is:

1. $\forall a, b, c, \qquad a \leq b \quad \Rightarrow \quad a + c \leq b + c$
2. $\forall a, b \ \text{ and } \ \forall c > 0, \qquad a \leq b \quad \Rightarrow \quad a \cdot c \leq b \cdot c$

- We observe that all the rules of algebraic computation follow from the properties: R_1, R_2, R_3.

[1] an algebraic structure is a set, called the underlying set (of the structure), equipped with one or more operations

Example

For example, all the usual procedures used to solve inequalities are a consequence of the axioms and of the algebraic properties of the sum and the product, expressed by R_1 and R_2, **and of the compatibility of the order relation with these operations**, expressed by R_3.

It is indeed R_3 that allows one to add the same number to both sides of an inequality and to multiply them by the same positive number without changing the direction of the inequality.

Ordered fields

Introduction: Ordered fields

- In this section we define **fields** and **ordered fields**
- We will see that $\mathbb{Q}$ and $\mathbb{R}$ are both ordered fields
- But $\mathbb{Q}$ is inadequate for expressing lengths: some "empty places" remain on the line

Ordered field

Definition 3: of (ordered) field

*A **field** is a set on which two operations (sum and product) are defined, satisfying the properties R_1 and R_2 and the **distributive property**.*
*An **ordered field** is a field on which a total order relation satisfying the property R_3 is also defined.*

- The distributive property is part of the definition of a field: it is the only property that links the sum and the product to each other.

$\mathbb{Q}$ and $\mathbb{R}$ are ordered fields

- Everything we have said so far holds both for the set of rational numbers and for the set of real numbers.

Observation 3

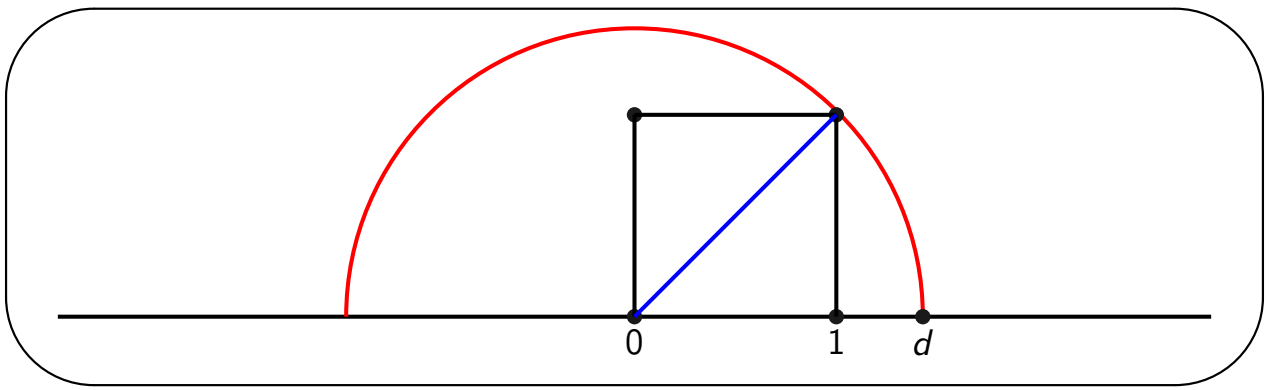
The sets of rational numbers $\mathbb{Q}$ and of real numbers $\mathbb{R}$ are ordered fields.

$\mathbb{Q}$ and $\mathbb{R}$

- From this point of view, therefore, $\mathbb{Q}$ and $\mathbb{R}$ appear to have similar properties. In what follows we will highlight the **property that substantially distinguishes** $\mathbb{R}$ from $\mathbb{Q}$ and that makes $\mathbb{R}$ the **right setting for developing mathematical analysis**.
- We begin by observing that the **set of rational numbers is inadequate for expressing the lengths of segments** (but also areas, volumes, times, speeds, etc.)

The diagonal of the square

As we have seen, there exist magnitudes that are not **commensurable** with each other. The classic example is given by the diagonal and the side of a square: if the side measures 1, the abscissa d that measures the diagonal is not a rational number.



The "empty places" on the line

Indeed, by the Pythagorean Theorem we have:

$$d^2 = 1^2 + 1^2 = 2$$

but, as we have seen, there is no rational number whose square is 2.

Hence the point d on the line does not represent any rational number. This means that, after "occupying" the points of the line with the rational numbers, there are still some **empty places** left on it.

Summary

- "$\leq$" is a total order relation on $\mathbb{Q}$ and on $\mathbb{R}$
- R_3: the order is compatible with sum and product (for $c > 0$)
- **Field**: R_1, R_2; **ordered field**: R_1, R_2, R_3
- $\mathbb{Q}$ and $\mathbb{R}$ are ordered fields
- The diagonal of the unit square has no rational measure: in $\mathbb{Q}$ the line has "empty places"

Bounded sets, maxima and minima

Introduction: Bounded sets, maxima and minima

- In this section we define **bounded** sets (above, below)
- We define the **maximum** and the **minimum** of a set
- With many examples we will see that a bounded set may have no maximum or no minimum

Bounded set

Definition 4: of bounded set

Let E be a set contained in $\mathbb{Q}$ or in $\mathbb{R}$. The set E is called ***bounded*** *if there exist two numbers m and M such that:*

$$\forall x \in E, \qquad m \leq x \leq M$$

- E is **bounded below** if there exists a number m such that: $\forall x \in E,\ m \leq x$
- E is **bounded above** if there exists a number M such that: $\forall x \in E,\ x \leq M$
- It makes no difference whether we require m and/or M to belong to $\mathbb{Q}$ or to $\mathbb{R}$, since the condition only concerns the existence of such numbers.

Why it makes no difference

This is not an immediate consequence of the definition, because $\mathbb{Q}$ is a proper subset of $\mathbb{R}$: it follows from the **Archimedean property** (recalled in the chapter *Sets* and proved later in this chapter).

- If $M \in \mathbb{R}$ satisfies $x \leq M$ for every $x \in E$, there exists a natural number $n > M$, and hence $x \leq M < n$ for every $x \in E$, with $n \in \mathbb{Q}$
- If $m \in \mathbb{R}$ satisfies $m \leq x$ for every $x \in E$, there exists a natural number $n > -m$, that is $-n < m$, and hence $-n < x$ for every $x \in E$, with $-n \in \mathbb{Q}$
- Conversely, every rational number is also a real number: the two requirements single out the same bounded sets

Maximum and minimum

Definition 5: of maximum and minimum of a set

An element $x_M \in E$ is called the ***maximum*** *of E if:*

$$\forall x \in E, \qquad x \leq x_M$$

An element $x_m \in E$ is called the ***minimum*** *of E if:*

$$\forall x \in E, \qquad x_m \leq x$$

- Note that the maximum and the minimum, by definition, **belong** to the set E

Uniqueness of the maximum and of the minimum

Observation 4

If a set E has a maximum, this is ***unique****; in the same way, if it has a minimum, this is unique. We then write* $\max E$ *and* $\min E$*.*

Proof

Let x_M and x'_M be two maxima of E:

- since x'_M belongs to E and x_M is a maximum of E, we have $x'_M \leq x_M$;
- since x_M belongs to E and x'_M is a maximum of E, we have $x_M \leq x'_M$.

From the antisymmetric property of the order relation it follows that:

$$x'_M \leq x_M \quad \text{and} \quad x_M \leq x'_M \quad \Longrightarrow \quad x_M = x'_M$$

The proof for the minimum is identical: if x_m and x'_m are two minima, then $x_m \leq x'_m$ and $x'_m \leq x_m$, and hence $x_m = x'_m$. □

Maximum, minimum and boundedness (1/2)

The existence of the maximum and of the minimum of a set implies that the set is bounded, that is:

$$\text{existence of maximum and minimum} \quad \Rightarrow \quad \text{bounded set} \tag{1}$$

but the converse is not true (we will give a counterexample later):

$$\text{bounded set} \quad \not\Rightarrow \quad \text{existence of maximum and minimum} \tag{2}$$

Hence the fact that a set is bounded is a necessary but not sufficient condition for the set to have a maximum and a minimum. Moreover, the existence of a maximum and a minimum is a sufficient but not necessary condition for a set to be bounded.

Maximum, minimum and boundedness (2/2)

From the **contrapositive** of (1) we have:

$$\text{unbounded set} \;\Rightarrow\; \text{the maximum does not exist } \mathbf{or} \text{ the minimum does}$$

that is, if a set is not bounded, it has no maximum or it has no minimum: it may well have one of the two, and indeed $\mathbb{N}$ is unbounded and nevertheless has minimum 0.

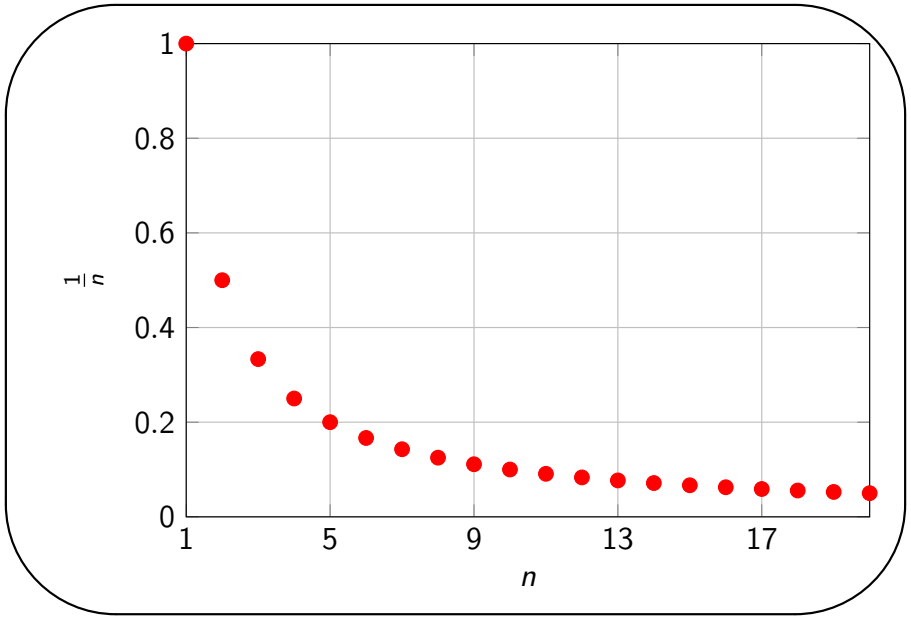
Example

Maxima and minima.

Example	set E	minimum	maximum
I)	$\mathbb{N}$	0	does not exist
II)	even integers	does not exist	does not exist

Example

Example	set E	minimum	maximum
III)	$\left\{ \frac{1}{n} \ : \ n \in \mathbb{N}_{>0} \right\}$	does not exist	1

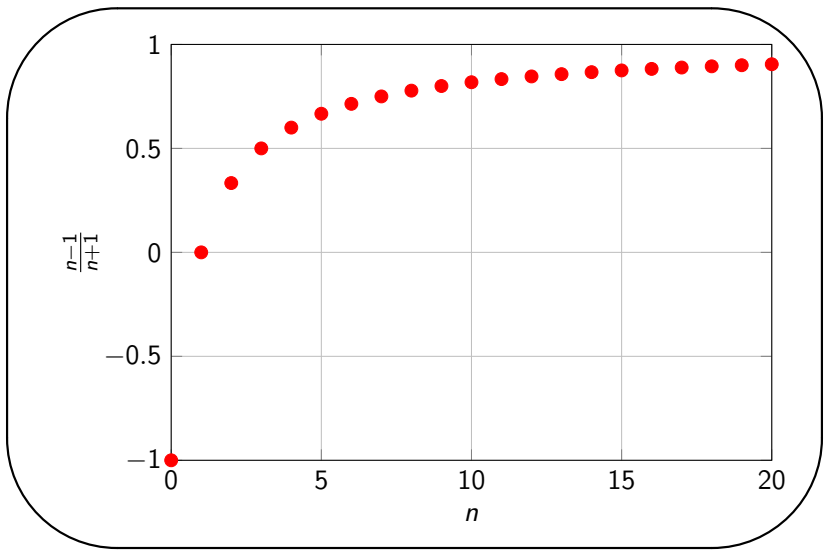


Example

Example	set E	minimum	maximum
IV)	$\left\{ \frac{n-1}{n+1} \ : \ n \in \mathbb{N} \right\}$	-1	does not exist

We have:

$$\frac{n-1}{n+1} = \frac{n+1-1-1}{n+1} = 1 - \frac{2}{n+1}$$



Bounded sets without maximum or minimum

Note that sometimes, even though the set is bounded, it may have no maximum or no minimum.

Example

Example	set E	minimum	maximum
V)	$\left\{x \in \mathbb{R} : \quad 27 \leq x^3\right\}$	3	does not exist

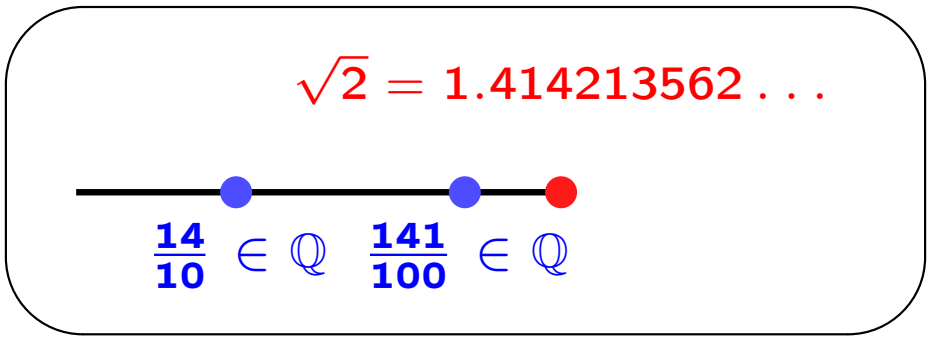
We have:

$$27 \leq x^3 \iff \sqrt[3]{27} = 3 \leq x \quad \text{and} \quad 3 \in E$$

Example

Example	set E	minimum	maximum
VI) $\{x \in \mathbb{Q} : \quad x \geq 0, x < \sqrt{2}\}$		0	does not exist

The value $\sqrt{2}$ is irrational, hence by truncating we obtain rational numbers that get closer and closer to $\sqrt{2}$, none of which has the property of being the maximum.

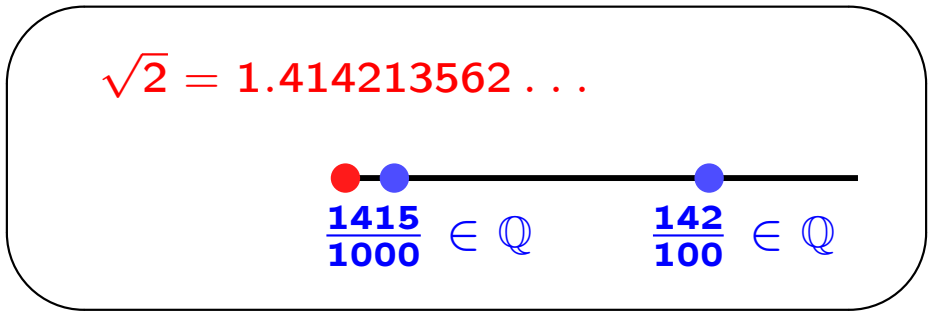


It is an example of a bounded set that has no maximum.

Example

Example	set E	minimum	maximum
VII) $\{x \in \mathbb{Q} : \quad x > \sqrt{2}, x \leq 4\}$		does not exist	4

The value $\sqrt{2}$ is irrational, hence by truncating and rounding up we obtain rational numbers that get closer and closer to $\sqrt{2}$, none of which has the property of being the minimum.



It is an example of a bounded set that has no minimum.

Example

Maxima and minima of intervals.

Example	set E	minimum	maximum
VIII)	$[-2, 1]$	-2	1
IX)	$(-\infty, 3)$	does not exist	does not exist

- In example VIII) the interval contains both of its endpoints: $-2 \in [-2, 1]$ and it is less than or equal to every $x \in [-2, 1]$; $1 \in [-2, 1]$ and it is greater than or equal to every $x \in [-2, 1]$
- In example IX) the interval is not bounded below, hence it has no minimum

Example

Why $(-\infty, 3)$ has no maximum. Given any $x \in (-\infty, 3)$, the number

$$y = \frac{x+3}{2} \qquad \text{satisfies} \qquad x < y < 3$$

- indeed $y - x = \frac{3-x}{2} > 0$ and $y < 3 \Leftrightarrow x < 3$
- hence $y \in (-\infty, 3)$ and it is strictly greater than x: no element of the set can be the maximum

Example

Maximum and minimum of a set defined by an inequality.

Example	set E	minimum	maximum
X)	$\{x \in \mathbb{R} : \quad -5 \leq 3x < 4\}$	$-\frac{5}{3}$	does not exist

It is convenient to **solve the inequality** first: dividing by $3 > 0$, and using the property R_3 which preserves the direction,

$$-5 \leq 3x < 4 \quad \Longleftrightarrow \quad -\frac{5}{3} \leq x < \frac{4}{3} \qquad \text{that is} \qquad E = \left[-\frac{5}{3}, \frac{4}{3}\right)$$

- $-\frac{5}{3} \in E$ and it is less than or equal to every element of E: it is the minimum
- The maximum does not exist: as in example IX), $y = \frac{1}{2}\left(x + \frac{4}{3}\right)$ satisfies $x < y < \frac{4}{3}$

Summary

- E **bounded**: there exist m, M with $m \leq x \leq M$ for every $x \in E$
- **Maximum** (minimum): element of E greater (less) than or equal to all the others
- Maximum and minimum $\Rightarrow$ bounded, but not conversely
- Examples: $\{\frac{1}{n}\}$ has no minimum; $\{\frac{n-1}{n+1}\}$ has no maximum
- In $\mathbb{Q}$: $\{x \geq 0,\ x < \sqrt{2}\}$ is bounded but has no maximum
- The maximum and the minimum, when they exist, are **unique**
- $[-2, 1]$ has both maximum and minimum; $(-\infty, 3)$ and $[-\frac{5}{3}, \frac{4}{3})$ have no maximum

Upper/lower bounds and suprema/infima

Introduction: Upper/lower bounds and suprema/infima

- In this section we define the **upper bounds** and **lower bounds** of a set
- The **supremum** (infimum) generalizes the maximum (minimum)
- We will see that in $\mathbb{Q}$ the supremum may not exist, while in $\mathbb{R}$ it exists

Upper and lower bounds

Definition 6: of upper bound of a set

*Let E be a set contained in $\mathbb{Q}$ or in $\mathbb{R}$; a number k belonging to $\mathbb{Q}$ or to $\mathbb{R}$ respectively is called an **upper bound** of E if:*

$$\forall x \in E, \qquad x \leq k$$

Definition 7: of lower bound of a set

*Let E be a set contained in $\mathbb{Q}$ or in $\mathbb{R}$; a number k belonging to $\mathbb{Q}$ or to $\mathbb{R}$ respectively is called a **lower bound** of E if:*

$$\forall x \in E, \qquad k \leq x$$

Upper and lower bounds: two remarks

We observe that a set bounded above (below) has many upper (lower) bounds. Moreover, the upper (lower) bounds do not necessarily belong to the set itself.

Observation 5

If k is an upper bound of E, then every number $h \geq k$ is still an upper bound of E; if k is a lower bound of E, then every number $h \leq k$ is still a lower bound of E. Hence the set of the upper (lower) bounds of E, if it is not empty, is a ***half-line****.*

Proof

Let k be an upper bound of E and let $h \geq k$. For every $x \in E$ we have $x \leq k$ by definition of upper bound, and hence, by the transitive property,

$$x \leq k \leq h \qquad \Longrightarrow \qquad x \leq h$$

that is, h is an upper bound of E.

The set of the upper bounds therefore contains, together with each of its elements k, all the numbers greater than k: it is thus a half-line unbounded above.

The proof for the lower bounds is analogous: if k is a lower bound of E and $h \leq k$, then for every $x \in E$ we have $h \leq k \leq x$ and hence $h \leq x$. □

Bounds that belong to the set

Observation 6

*An upper bound of E that **belongs** to E is the maximum of E; a lower bound of E that belongs to E is the minimum of E.*

Proof. Let k be an upper bound of E with $k \in E$. By definition of upper bound we have $x \leq k$ for every $x \in E$: these are exactly the two conditions that define the maximum of E, and hence $k = \max E$. In the same way, if k is a lower bound of E and $k \in E$, then $k = \min E$. □

Supremum and infimum

Definition 8: of supremum

*The **supremum** of E (denoted by $\sup E$) is the minimum of the upper bounds of E.*

Definition 9: of infimum

*The **infimum** of E (denoted by $\inf E$) is the maximum of the lower bounds of E.*

Suprema, maxima and minima

We observe that the supremum (infimum) may not exist (unbounded sets).

We also observe that if the set has a maximum (minimum), it coincides with the supremum (infimum).

The convention $\sup E = +\infty$

For unbounded sets the following **convention** is adopted: if E is not bounded above one writes

$$\sup E = +\infty$$

and if E is not bounded below one writes

$$\inf E = -\infty$$

- The symbols $+\infty$ and $-\infty$ **are not numbers**: this notation is only a compact way of saying that E has no upper (lower) bounds

Example (1/2)

Suprema and infima.

Example	set E	inf E	sup E	minimum	maximum
I)	$\{x \in \mathbb{Q} : \ x \geq 0, x < \sqrt{2}\}$	0	does not exist in $\mathbb{Q}$	0	does not exist

Here the upper bounds are looked for **inside** $\mathbb{Q}$. We have:

$$\sqrt{2} \notin E \quad \text{and} \quad \sqrt{2} \notin \mathbb{Q}$$

hence the maximum does not exist; and the sup does not exist **in** $\mathbb{Q}$ either, because the rational upper bounds of E are the rational numbers greater than $\sqrt{2}$ and among them there is no smallest one.

Example (2/2)

Example	set E	inf E	sup E	minimum	maximum
II)	$\{x \in \mathbb{R} : \ x \geq 0, \ x < \sqrt{2}\}$	0	$\sqrt{2}$	0	does not exist

We have:

$$\sqrt{2} \notin E \quad \text{and} \quad \sqrt{2} \in \mathbb{R}$$

that is, the maximum does not exist, but $\sqrt{2}$ is the minimum of the upper bounds and therefore it is the sup.

Example

Suprema and infima of intervals.

Example	set E	$\inf E$	$\sup E$	minimum	maximum
VIII)	$[-2,1]$	-2	1	-2	1
IX)	$(-\infty,3)$	$-\infty$	3	does not exist	does not exist
X)	$\{x \in \mathbb{R} : \; -5 \leq 3x < 4\}$	$-\frac{5}{3}$	$\frac{4}{3}$	$-\frac{5}{3}$	does not exist

- In examples VIII) and X) the minimum exists and therefore coincides with the infimum; in example VIII) the maximum exists as well and coincides with the supremum
- In example IX) the set is not bounded below, hence $\inf E = -\infty$ by the convention just introduced

Example

Why $\sup(-\infty, 3) = 3$.

- The number 3 is an upper bound, because $x < 3$ for every $x \in E$
- No number $k < 3$ is an upper bound, because $y = \frac{k+3}{2}$ satisfies $k < y < 3$ and therefore belongs to E
- Hence 3 is the minimum of the upper bounds, that is $\sup E = 3$, and it is not a maximum because $3 \notin E$

With the same argument, for $E = \left[-\frac{5}{3}, \frac{4}{3}\right)$ we obtain $\sup E = \frac{4}{3} \notin E$.

Example

Suprema and infima of $E = \left\{ \frac{1}{n} : n \in \mathbb{N}_{>0} \right\}$ (example III)): we have $\inf E = 0$ and $\sup E = \max E = 1$.

- 0 is a lower bound, because $\frac{1}{n} > 0$ for every $n \in \mathbb{N}_{>0}$
- no number $k > 0$ is a lower bound: by the **Archimedean property** there exists $n \in \mathbb{N}_{>0}$ with $n > \frac{1}{k}$, and hence $\frac{1}{n} < k$
- hence 0 is the maximum of the lower bounds, but it is not the minimum because $0 \notin E$

Summary

- **Upper bound** (lower bound): k with $x \leq k$ ($k \leq x$) for every $x \in E$
- $\sup E$: minimum of the upper bounds; $\inf E$: maximum of the lower bounds
- If the maximum (minimum) exists, it coincides with sup (inf)
- $\{x \geq 0,\ x < \sqrt{2}\}$: in $\mathbb{Q}$ it has no sup, in $\mathbb{R}$ it has sup $= \sqrt{2}$
- An upper bound that belongs to E is the maximum; the set of the upper bounds is a half-line
- Convention: $\sup E = +\infty$ ($\inf E = -\infty$) if E is not bounded above (below)

Axiomatic definition of the real numbers

Introduction: Axiomatic definition of the real numbers

- In this section we introduce the **supremum property** (R_4)
- It is the property that distinguishes $\mathbb{R}$ from $\mathbb{Q}$
- We will define $\mathbb{R}$ axiomatically and see the equivalent form with **cuts** (Dedekind axiom)

Supremum property

A totally ordered number set X is said to have the **supremum property** (least upper bound property) if:

$R_4 \rightarrow$ Every non-empty set $E \subset X$ that is bounded above has a supremum in X

- It is easy to prove that, if this property holds, then it is also true that every non-empty subset of X that is bounded below has an infimum.
- It is not required that X itself has a supremum (for example, in the case $X = \mathbb{Q}$ or $\mathbb{R}$ this is certainly false!), but that every non-empty subset of X that is bounded above has one.

$\mathbb{Q}$ and the supremum property

Example VI shows that $\mathbb{Q}$ certainly does not have the supremum property. On the other hand, $\mathbb{R}$ has this property.[2]

- In the **axiomatic definition** of $\mathbb{R}$, this property is part of the very definition of $\mathbb{R}$

Definition 10: (axiomatic) of the real numbers

We call $\mathbb{R}$ a set that satisfies the properties R_1, R_2 (together with the distributive property), R_3 and R_4, that is, an ordered field that has the supremum property

[2] However, we will not present a formal proof.

The Archimedean property

Proposition 1: Archimedean property

For every real number x there exists a natural number n such that:

$$n > x$$

- This is the property that we recalled in the chapter *Sets* and used above
- **It is not an additional axiom**: it follows from the property R_4

Proof

Suppose by contradiction that there exists a real number x such that $n \leq x$ for every $n \in \mathbb{N}$. Then $\mathbb{N}$ is a non-empty subset of $\mathbb{R}$ bounded above (by x), and therefore, by the property R_4, it has a supremum in $\mathbb{R}$:

$$s = \sup \mathbb{N}$$

Since s is the **minimum** of the upper bounds of $\mathbb{N}$, the number $s - 1$, smaller than s, is not an upper bound of $\mathbb{N}$: there exists therefore $n \in \mathbb{N}$ such that

$$n > s - 1 \qquad \Longrightarrow \qquad n + 1 > s$$

But $n + 1$ is still a natural number, and this contradicts the fact that s is an upper bound of $\mathbb{N}$. □

An equivalent form

For every real number $\varepsilon > 0$ there exists $n \in \mathbb{N}_{>0}$ such that $\frac{1}{n} < \varepsilon$.

- It is enough to choose $n > \frac{1}{\varepsilon}$, and then $\frac{1}{n} < \varepsilon$
- As a consequence, the only non-negative number that is less than or equal to $\frac{1}{n}$ for every $n \in \mathbb{N}_{>0}$ is zero

Dedekind axiom

- The supremum property is also called the **Dedekind axiom**, **or axiom of continuity**, or **completeness axiom**, and it can also be stated in the following equivalent form.

Let $\{A, B\}$ be a partition of $\mathbb{R}$ (i.e., A and B are non-empty disjoint sets whose union is $\mathbb{R}$); the partition is called a **cut** (section) if:

$$\forall a \in A \quad \text{and} \quad \forall b \in B \quad \text{we have} \quad a < b.$$

Separating element

Then one proves that:

$R_4' \rightarrow$ For every cut $\{A, B\}$ of $\mathbb{R}$ there exists a unique real number s (called the **separating element**) such that:

$$\forall a \in A \quad \text{and} \quad \forall b \in B, \qquad a \leq s \leq b$$

This separating element is $\sup A$ and $\inf B$, hence $\sup A = \inf B$

Summary

- R_4: every non-empty set bounded above has a supremum
- $\mathbb{Q}$ does not have this property, $\mathbb{R}$ does
- $\mathbb{R}$ is an ordered field with the supremum property (R_1–R_4)
- Equivalent form (Dedekind): every cut $\{A, B\}$ has a unique separating element $s = \sup A = \inf B$
- The **Archimedean property** is not an additional axiom: it follows from R_4

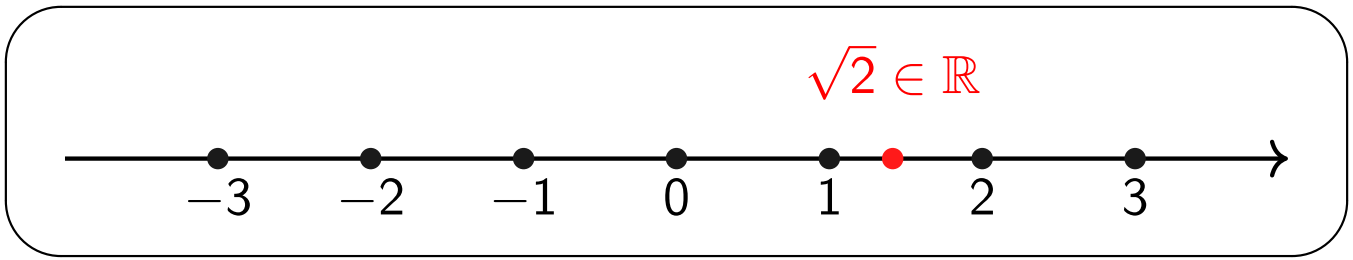
Geometric representation of $\mathbb{R}$

Introduction: Geometric representation of $\mathbb{R}$

Can the set of real numbers $\mathbb{R}$ be put in one-to-one correspondence with the points of the Euclidean line?

The real line

- Let us now return to the problem of the incommensurability of the side and the diagonal of a square. When, with straightedge and compass, having fixed the unit side of the square, we construct its diagonal and transfer it onto the starting line, the arc drawn by the compass "splits the line in two", generating what we have called a cut; the separating element, which exists by axiom R_4, geometrically represents the point of intersection, and therefore that number measures the diagonal.



The real line

Hence the geometric interpretation of property R_4 shows us how the set $\mathbb{R}$ is an adequate representation of our intuitive idea of a line, so adequate that **in mathematics the expression "the real line" is often used to denote** $\mathbb{R}$, identifying the number set with its natural geometric representation.

Summary (1/2)

Algebraic structure and order:

- R_1, R_2: properties of sum and product (field)
- R_3: total order compatible with the operations (ordered field)
- $\mathbb{Q}$ and $\mathbb{R}$ are both ordered fields

Limits of $\mathbb{Q}$:

- There exist incommensurable magnitudes (diagonal and side of a square)
- In $\mathbb{Q}$ bounded sets may have no supremum

Summary (2/2)

Suprema and infima:

- Bounded sets, maxima and minima, upper and lower bounds
- $\sup E$ = minimum of the upper bounds, $\inf E$ = maximum of the lower bounds

Real numbers:

- $\mathbb{R}$ is the ordered field with the supremum property (R_4)
- Equivalent form: Dedekind axiom (cuts and separating element)
- From R_4 follows the **Archimedean property**: for every $x \in \mathbb{R}$ there exists $n \in \mathbb{N}$ with $n > x$
- $\mathbb{R}$ faithfully represents the line: "the real line"